Mayur Gadekar

AI/ML Engineer

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PROFILE

AI/ML Engineer building production GenAI systems for enterprise automation, including agentic workflows, RAG, multilingual voice agents, and document intelligence. Delivered measurable impact: 90%+ improvement in response accuracy and 85% less manual document processing. M.Tech candidate in Applied AI at VNIT Nagpur, with research in multimodal vision-language systems and reinforcement learning.

WORK EXPERIENCE

Aug 2025 – Present

AI/ML Engineer, Imperative Business Ventures

- Built LangGraph RAG apps with Gemini Live, FAISS, MySQL, and MongoDB, improving accuracy by 90%.
- Built FastAPI services and WebSocket/Plivo voice workflows, improving customer response time by 60% vs. manual handling.
- Integrated document intelligence and liveness checks, reducing document-processing time by 85%.

Feb 2025 – Aug 2025

AI/ML Intern, Imperative Business Ventures

- Built Python/FastAPI AI apps with MySQL and MongoDB, streamlining backend workflows and speeding AI-feature integration.
- Developed FAISS RAG pipelines; internal tests estimated 15–20% fewer incorrect or inconsistent LLM responses.
- Implemented structured outputs, function calling, and validation to improve LLM response consistency.

Oct 2024 – Jun 2025

AI Prompt Engineer, Outlier

- Integrated GPT-4o, Llama, and Mistral into NLP workflows, using optimized prompts to cut inconsistent or problematic outputs by 10–15% in internal tests.
- Benchmarked model versions and evaluated safety to identify and address problematic responses.

Jan 2023 – May 2023

Research Analyst Intern, IGurus Consultancy Services LLP (ICS)

- Developed and optimized a Hindi transliteration module using NLP, improving conversion consistency and reducing errors through systematic testing.

Jan 2023 – Feb 2023

Web Developer Intern, Aller Technologies

- Built a Society Management System with Flask/Django and React.js, digitizing routine community workflows to reduce manual coordination and improve resident-management communication.

EDUCATION

Aug 2024 – Sep 2026

M.Tech (Applied Artificial Intelligence), Visvesvaraya National Institute of Technology (VNIT) · Nagpur

Completed September 2026

Aug 2020 – Jun 2024

B.E. (Artificial Intelligence and Data Science), Ajeenkya DY Patil University · Pune

PROJECTS

Jan 2025 – Present

Vendor Comparison AI Platform

- Designed and developed an enterprise AI-powered tender analysis platform for processing technical and commercial procurement documents at scale.
- Built BOQ-driven comparison pipelines using FastAPI, Gemini, FAISS, and semantic retrieval for automated vendor evaluation and scoring.
- Implemented AI-powered vendor scoring, technical compliance validation, and commercial benchmarking using RAG-based document intelligence workflows.
- Reduced manual tender comparison effort by over 80%, enabling faster and more accurate procurement decision-making.

Feb 2025 – Present

AI Voice Agent — Multilingual Inbound Call Automation

- Built production-grade multilingual AI voice agents using Gemini Live API, FastAPI, WebSockets, and Plivo telephony for real-time enterprise customer support automation.
- Implemented real-time speech-to-text processing, LLM orchestration, tool/function calling, and dynamic conversation management with stateful agent memory.
- Developed context-aware identity verification workflows and multi-turn slot capture, maintaining session continuity across complex multi-step conversation flows.
- Optimized end-to-end latency and response quality, improving customer response time by 60% compared to manual handling benchmarks.

Jan 2025 – Jan 2026

Self-Improving Vision-Language Support System

- Designed a memory-augmented multimodal AI system combining BLIP-2 for vision-language understanding with PPO-based reinforcement learning for iterative self-improvement, analogous to RLHF alignment in production LLM systems.
- Built a retrieval-augmented memory architecture using FAISS for episodic cross-session retrieval, enabling grounded and contextually consistent agent responses.
- Achieved 73.8% accuracy on VQAv2 and 68.3% on GQA benchmarks through reward-shaped PPO training; published in IEEE conference format.

Jan 2025 – Present

CSAT/NPS Automation Platform

- Developed an automated customer satisfaction and NPS survey platform using FastAPI, React, and MySQL with multi-channel distribution (email, WhatsApp, SMS), processing 10,000+ survey interactions.
- Integrated HuggingFace transformer models — RoBERTa for sentiment analysis and BART for zero-shot categorization — with real-time dashboards, improving actionable CX insight delivery by 60%.

Jan 2025 – Present

DocExtract India — AI Document Intelligence

- Architected a multimodal document intelligence pipeline using LayoutLMv3 and Tesseract OCR for structured extraction from Indian government documents (Aadhaar, PAN, passports), achieving 95% extraction accuracy.
- Built a production-ready FastAPI backend with Pydantic v2 validation, robust error handling, and optimized API pipelines, reducing document processing time by 85%.

SKILLS

LLM & Generative AI

LangChain • LangGraph • Gemini API • OpenAI (GPT-4o) • HuggingFace Transformers • Llama • Mistral • Prompt Engineering • Instruction Tuning • Chain-of-Thought Reasoning • Function/Tool Calling • Agentic Orchestration • RAG Pipelines • Graph RAG • Semantic Search • Agent Memory Systems • OpenAI API • Anthropic API • Google Gemini API • LlamaIndex • Structured LLM Outputs • LLM Agent Frameworks

Multimodal & Deep Learning	BLIP-2 • LayoutLMv3 • CLIP • RoBERTa • BART • Whisper • PyTorch • TensorFlow • Scikit-learn • OpenCV • PPO (Reinforcement Learning) • RLHF • PEFT • LoRA
Vector Databases & Retrieval	FAISS • ChromaDB • pgvector • Pinecone • Embedding Pipelines • Semantic Search • Vector Databases
LLMOps & MLOps	MLflow • Prompt Evaluation • Model Monitoring • Model Versioning • Experiment Tracking • Docker • CI/CD • API Deployment
Cloud & Infrastructure	AWS (EC2, S3) • GCP (Vertex AI) • Docker • Linux • REST APIs • Amazon Web Services • Google Cloud Platform • Microsoft Azure
Backend & APIs	Python (Experienced) • FastAPI • Flask • Django • Async Programming • WebSockets • Microservices • Python
Voice AI & NLP	Plivo Telephony • Real-time STT • Sentiment Analysis • Zero-shot Classification • NER • Multilingual NLP
Databases	MySQL • PostgreSQL • FAISS • ChromaDB • pgvector • MongoDB • Redis
Other	Git • Postman • Power BI • Agile Methodologies • JavaScript • SQL • C++ • HTML • CSS • React.js • Software Testing • Code Review Practices
Generative AI Expertise	Multi-Agent Systems (LangGraph, CrewAI, AutoGen) • Retrieval-Augmented Generation (RAG & Graph RAG) • Vector Search & Embeddings • AI Voice Agents & Conversational AI • Function Calling & Tool Use • Prompt Engineering & LLM Evaluation • AI Workflow Automation • Agent Memory Systems • Semantic Search • LLMOps & Model Monitoring • Retrieval-Augmented Generation • Function Calling for LLMs

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COURSES

Present	**Certification in Machine Learning and Python Programming**
Present	**Microsoft Power BI - Up & Running with Power BI Desktop**
Present	**Artificial Neural Networks (ANN) With Keras in Python**
Present	**Advanced: Generative AI for Developers Learning Path**